

The Pile Game

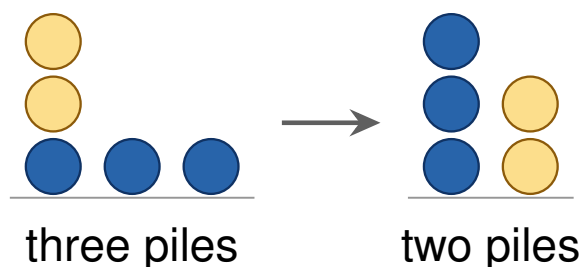
Week 1 · K–1 table

Name: _____

Grown-up: read the words on these pages aloud. The children do every move with real counters, then draw what they see.

The rule

Take one counter from **every** pile. Put those counters together. They make a **new pile**. If a pile had only one counter, that pile is **gone**.



Look at the picture. There are three piles. The dark counters are the ones we take, one from every pile. They make a new pile of three (dark). The big pile has two counters left. The two little piles are gone.

Your turn. Make piles of 3, 1 and 1 with your counters. Do the move. Do you get a pile of 3 and a pile of 2?

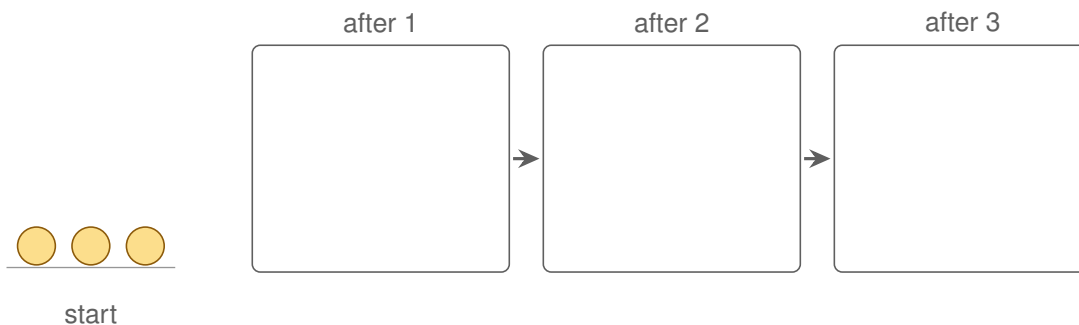
Again. Make piles of 2, 2 and 2. Do the move. Draw the piles you get.



How many piles do you have now? Which pile is the new one?

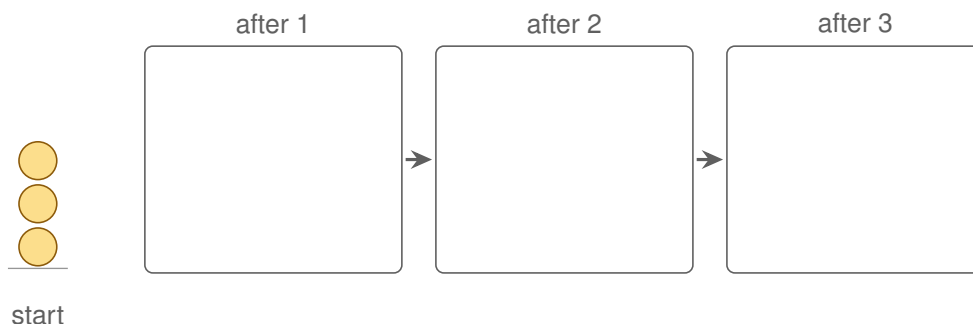
Three counters

Put three counters in three piles of one, like this. Do the move. Draw the piles you get. Do it again. And again.



Do the piles stop changing? When the piles stay the same after a move, we say the game is **STUCK**.

Now put the three counters in **one** pile. Do the move again and again.



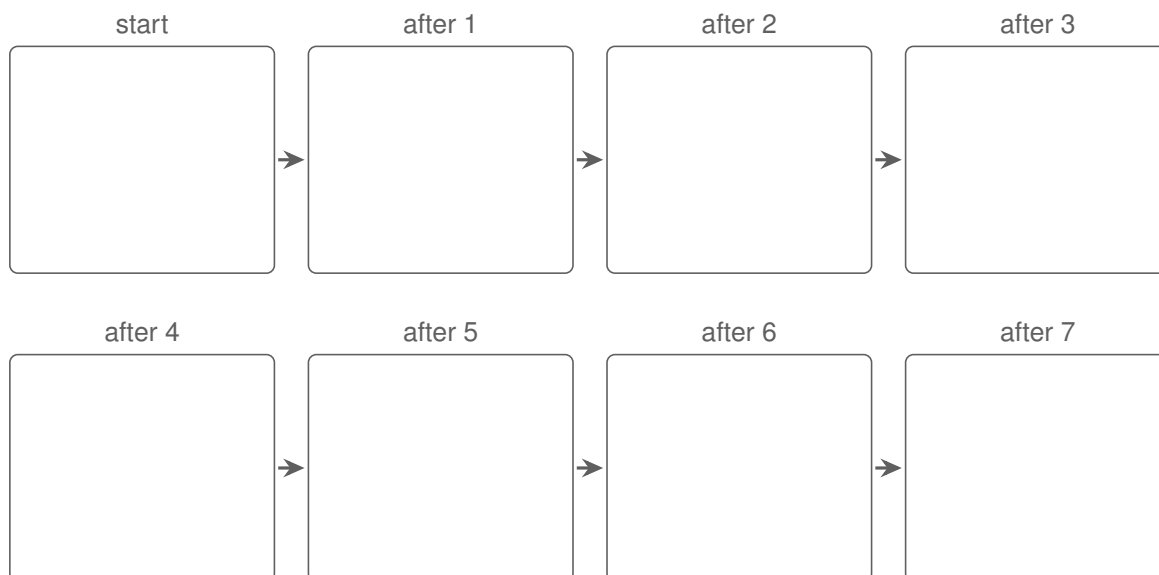
Did you get stuck at the **same** piles as before? Compare with a friend.

A large empty rectangular box with a thin black border, intended for drawing the state of the game when it gets stuck.

Draw the stuck piles here, nice and big.

Six counters

Take six counters. Make piles any way you like. Draw your start. Then do the move again and again. Draw the piles every time.



Did the game get stuck? Draw the stuck piles big.

A large empty rectangular box for drawing stuck piles.

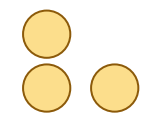
Everyone at the table started with different piles. Did everyone get stuck at the **same** piles?

Staircases

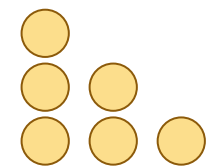
These piles are called **staircases**. Each pile is one step shorter than the pile before it.



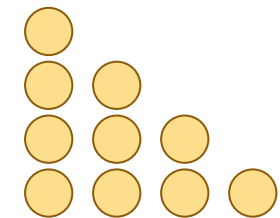
1 counter



3 counters



6 counters

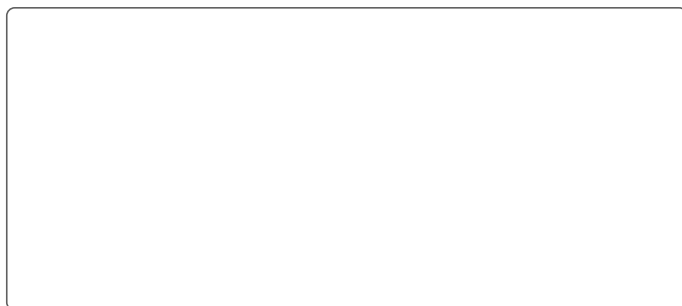


counters

Count the counters in the biggest staircase. Write the number in the box.

Your six counters got stuck at a staircase! Point to it.

Build the **next** staircase with your counters. It has five piles. Draw it here. How many counters did you use?

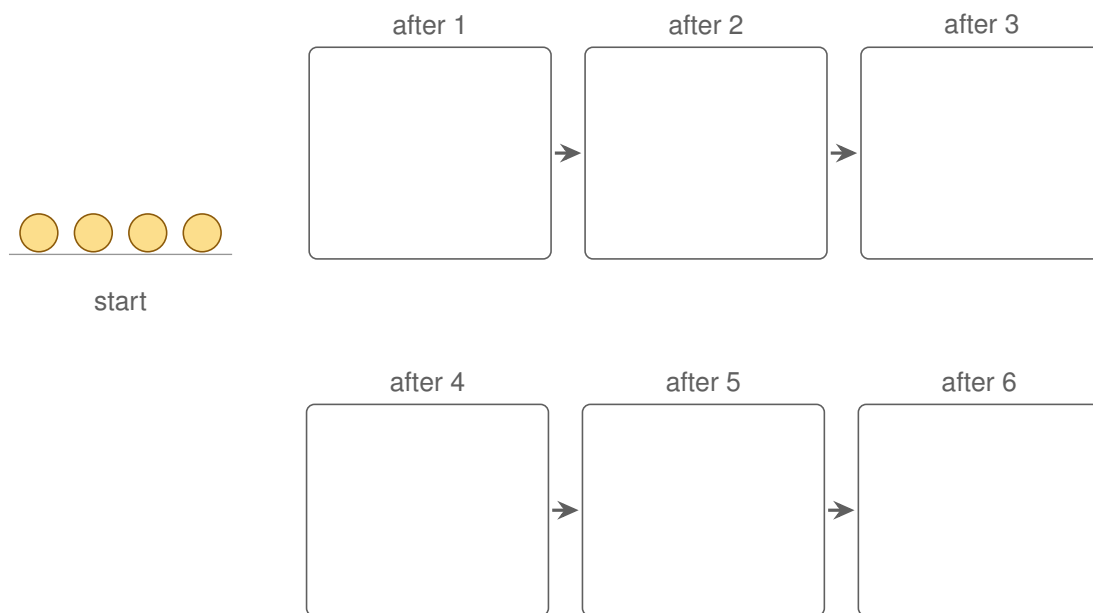


counters

Can you build a staircase with **8** counters? Try it! What goes wrong?

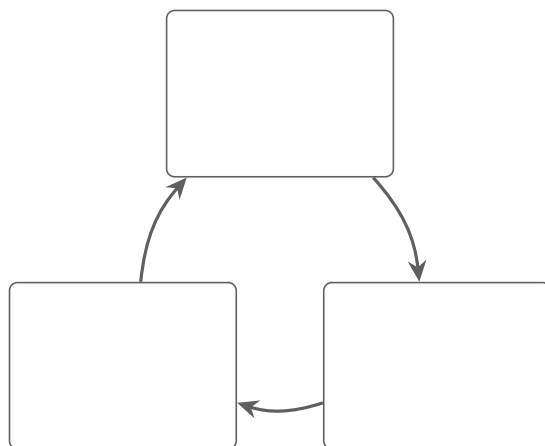
Four counters

Take just four counters. Start with four piles of one. Do the move again and again. Draw the piles every time.



Does the game get stuck? Circle one: **YES** **NO**

Three kinds of piles come back again and again, like a merry-go-round. Draw the three of them, in order, and follow the arrows.



More to try

Ten counters. Play with your whole table. Where do you get stuck? Draw it. Is it a staircase?



Five counters. Start any way you like. Stuck, or merry-go-round? Draw what comes back again and again.



Fifteen counters. Can you build a staircase with fifteen counters? How many piles does it have?



Cards. Put a card next to each pile that shows how many counters are in it. After a move, do you need new cards?